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LAB V

General Objectives:

The Software Engineering Lab has been developed by keeping in mind the following objectives:

- To impart state-of-the-art knowledge on Software Engineering and UML in an interactive manner through the Web.
- Present case studies to demonstrate practical applications of different concepts.
- Provide a scope to students where they can solve small, real-life problems.

Lab Objectives:

By the end of this lab, students will be able to:

- Clearly list functional requirements in standardized form.
- Identify and correct poorly written or “bloated” requirements.
- Group related requirements into meaningful responsibility categories.
- Prepare the foundation for modular decomposition (next lecture).

Precautions:

- Use previous knowledge from SWE202 and other online sources except an AI chatbot.
- **Do not copy your friend’s work, and avoid direct AI insertion.**
- Use actual collaborative tools/frameworks/IDEs where necessary. You may use Java, Python, or any other object-oriented language, but your code should follow object-oriented principles as shown in your UML diagram
- Use this exact manual, At the end of the day, convert it into a **PDF** for the final submission.
- Answer each question on a **fresh page**.

The Scenario (Note that this scenario may be used for the entire Lab:

Near East University Hospital wants to implement a Remote Health Monitoring System to track patients with chronic conditions such as hypertension and diabetes. Patients use wearable devices to record their vital signs (heart rate, blood pressure, temperature, etc.), and the data is automatically sent to the hospital’s system. If the system detects abnormal values, it generates an alert and forwards it to the doctor for review. Doctors can then provide recommendations or schedule follow-ups with the patients.

Instructions:

1. Rewrite and List Functional Requirements Clearly
2. Identify Bloated or Combined Requirements
3. Group Requirements into High-Level Responsibility Areas
4. Describe Why Grouping Requirements Matters for Architecture
5. What are the differences between requirements and features?

1 FUNCTIONAL REQUIREMENT

The functional requirement for the remote monitoring system will provide details of what the system does. This includes:

User Authentication: The system will authenticate patient and doctor using secure log-in credentials.

Collecting Vitals: The system will collect patient vitals which includes temperature, heart rate, and blood pressure.

Detect abnormality: The system will check for abnormal readings

Transmit data: The The system will transmit data from the IoT device to the server.

Generate alert: The system will generate alert when abnormal readings are detected

Doctor Recommendation: The system will send doctor's recommendation to the patient

Health data storage: The system will store patient's health data in a database

Group Requirements into High-Level Responsibility Areas

User Management model

User authentication

Collect vitals

Receive feedback and recommendation

Review alert

Data management model

Receive vitals

Analyze vitals

Send vitals

Store data

Update data

Notification management model

Send alert(if abnormality is detected)

Send feedback and recommendation

Why Grouping Requirements Matters for Architecture

The reason why grouping requirement matters in architecture includes:

To define the system layer: Each layer perform different function, by grouping the requirements, we can easily understand which layer is doing what.

To prevent bloating: To clearly explain what each layer responsible for, this eliminate duplication

To enable scalability: By grouping the requirements, we can easily scale each layer independently without derupting the entire system

To enable maintainability: This means we can upgrade, or update for example, the user interface layer without changing the business logic

What are the differences between requirements and features

- The requirements are what the system must do this include:
- The system shall collect the patient vitals from the IoT device
- The system shall analyze data to detect abnormal readings
- The system shall notify the doctor
- The system shall store patient record

The featured requirements focuses on the user interaction with the system. In

the remote health monitoring system, the featured requirements include:

- Real time patient monitoring
- Automatic alert generation for the doctor
- Continuous tracking of patient's vitals